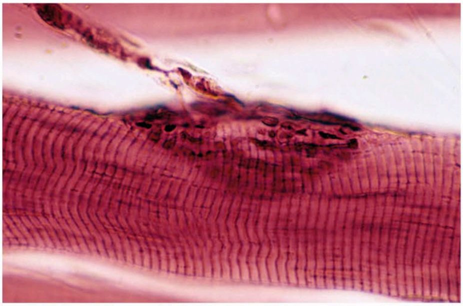
MUSCLES OF THE FACE, HEAD, AND NECK

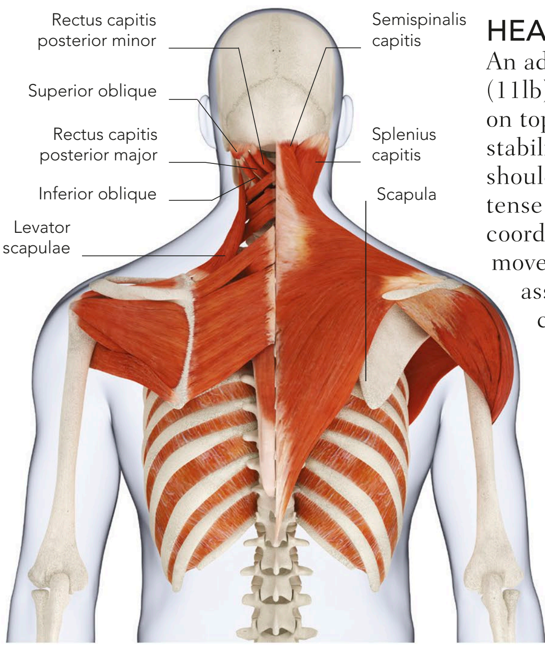
THE MUSCLES OF THE FACE, HEAD, AND NECK INTERACT TO STEADY AND MOVE THE HEAD AND TO MOVE THE FACIAL FEATURES. THE MUSCULATURE INVOLVED IS HIGHLY COMPLEX, MAKING POSSIBLE A HUGE RANGE OF FACIAL EXPRESSIONS.



NERVE–MUSCLE JUNCTION
In this microscope image, a nerve cell (top left) joins a facial muscle fibre. At the point of contact between the two is the motor end plate (centre), an area of highly excitable muscle fibre.

FACIAL MUSCLES

Some facial muscles are anchored to bones. Others are joined to tendons or to dense, sheet-like clusters of fibrous connective tissue called aponeuroses. This means that some facial muscles are joined to each other. Many of these muscles have their other end inserted into deeper layers of the skin. The advantage of this complex system is that even a slight degree of muscle contraction produces movement of the facial skin, which reveals itself as a show of expression or emotion. Almost all facial muscles are controlled by the facial nerve called cranial VII (see p.102).



HEAD AND NECK MUSCLES

An adult's head weighs more than 5kg (11lb) and is, to some extent, "balanced" on top of the vertebral column. Strong, stabilizing muscles in the neck, inner shoulders, and upper back constantly tense to steady the head and contract in coordinated teams to produce complex movements of the neck. These muscles assist facial expressions and non-verbal communication, such as emphasizing doubt by cocking the head slightly to one side, or moving the head to indicate "yes" or "no".

BACK MUSCLES
The neck and shoulder muscles support and steady the head. Upper-back muscles that attach to the shoulder blade (scapula) help to stabilize the shoulders.